CASE STUDY AND SOFTWARE REQUIREMENTS SPECIFICATION

Donation and Relief Fund Management System

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1.0	23-Jul-2026	Swara Padole	Initial formal release of the Software Requirements Specification.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Donation and Relief Fund Management System. It is the primary reference for the design, development, testing, and evaluation of the system, and it establishes a shared, unambiguous understanding among all stakeholders of what the system will do and the standards it must meet.

1.2 Scope

This document covers a web-based application through which individual donors and relief organizations (NGOs/federations) may register, discover, fund, and transparently track disaster-relief campaigns using blockchain-backed smart contracts. It defines the functional behaviour of registration, campaign discovery, secure authentication, fund transfer, ledger visibility, and fund withdrawal, together with the non-functional standards of availability, throughput, security, and cross-platform compatibility the system must satisfy.

The internal governance policies of individual relief organizations and the specific consensus mechanism of the underlying blockchain network fall outside the scope of this document and are treated as external dependencies.

1.3 Intended Audience

This document is intended for the development team, project reviewers and academic evaluators, and any future maintainers of the system. Readers are assumed to have a basic understanding of web application architecture and foundational blockchain concepts.

1.4 Definitions, Acronyms and Abbreviations

The following terms and abbreviations are used consistently throughout this document.

Term	Definition
SRS	Software Requirements Specification — this document.
NGO	Non-Governmental Organization; used interchangeably with "Federation" to describe a relief organization.
FR / NFR	Functional Requirement / Non-Functional Requirement.
DLT	Distributed Ledger Technology; the general class of technology underlying the blockchain layer.
Smart Contract	Self-executing code deployed on the blockchain that automatically enforces the terms of a transaction without an intermediary.
Wallet Address	A unique cryptographic identifier used to send, receive, and route funds on the blockchain network.
Ledger	The append-only, publicly verifiable record of every transaction processed by the system.
Uptime	The proportion of time the system remains operational and accessible to users.

Term	Definition
SHA-256	A cryptographic hash function used to irreversibly transform sensitive data, such as passwords, into a fixed-length string for secure storage.

1.5 References

No external standards documents were used verbatim in the preparation of this specification. Where applicable, the structure and section conventions of this document follow the general spirit of the IEEE 830 Software Requirements Specification standard, adapted for academic case-study purposes.

2. Overall Description

2.1 Product Perspective

The Donation and Relief Fund Management System is a new, self-contained product rather than a component of an existing system. It integrates a conventional web front end with a blockchain-based back end, positioning the smart contract layer as the authoritative source of truth for all financial transactions while the web layer serves purely as a presentation and orchestration layer.

2.2 User Classes and Characteristics

- **Donor:** A registered individual who contributes funds to campaigns. Assumed to have minimal technical expertise and to interact primarily through simplified, e-commerce-style flows.
- **Organization / Federation:** A verified NGO or relief body that publishes campaigns and withdraws disbursed funds. Subject to additional verification given its elevated financial responsibility.
- **Visitor (unregistered):** Any member of the public who may browse and search campaigns without an account, evaluating the system's transparency before choosing to register.
- **System Administrator:** An implicit oversight role responsible for organization verification. Administrators cannot modify, delete, or obscure ledger records once written, consistent with the immutability requirement.

2.3 Operating Environment

The system is delivered as a browser-based web application requiring no local installation. It must operate consistently across Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge, on both desktop and mobile viewports, while communicating with a blockchain network for all transaction-related operations.

2.4 Assumptions and Dependencies

- **Connectivity:** Donors and organizations are assumed to have access to a stable internet connection.
- **Network Reliability:** The system's own 99.9% availability target depends on the uptime and consensus guarantees of the underlying blockchain network, which are outside this system's direct control.
- **Organization Vetting:** Organizations are assumed to be vetted for legitimacy before being granted campaign-publishing and withdrawal rights; this verification workflow is treated as a dependency rather than a feature defined in this document.
- **Key Custody:** Users are assumed to retain custody of their own private wallet credentials, as the system explicitly does not store them in plain text and takes no custodial responsibility for lost keys.

3. System Overview and Problem Statement

In the modern landscape of philanthropy and crisis management, the primary bottleneck in disaster relief is rarely a lack of capital; it is a deficit of trust. When humanitarian crises occur, prospective donors are often deterred by opaque fund routing, administrative overhead, and the fear of misallocation.

To resolve this, it is proposed to develop the Donation and Relief Fund Management System. The core architectural philosophy of this system is to maintain a simple, accessible user interface while utilizing a robust, immutable blockchain network as its backend infrastructure. This hybrid approach ensures a transparent ledger in which every transaction between a donor and a receiving organization (or federation) is permanently recorded, publicly verifiable, and impossible to alter.

The final deliverable will be a web-based application. While the front end will remain lightweight and user-friendly, the back end will leverage smart contracts to guarantee impartial fund distribution. Strict security measures will be enforced, ensuring no sensitive user data or wallet credentials are stored in plain text.

4. Functional Requirements

Based on the architectural goals of transparency and simplicity, the core functionalities of the system are identified below. Each requirement is expressed with its responsible actor(s), a description of expected behaviour, and the pre- and post-conditions that govern it.

FR1 — New User Registration

Actor(s)	Donor, Organization (Federation/NGO)
Description	The system accommodates two distinct user roles: Donors and Organizations. Any entity wishing to participate must register. Upon successful registration, the user receives login credentials and a unique blockchain wallet address is linked to their profile for transaction routing.
Pre-condition	The user is not already registered in the system.
Post-condition	A user profile is created with valid credentials and a linked wallet address, enabling participation in donations or campaign management.

FR2 — Search and Verify Campaigns

Actor(s)	Donor, Organization, Visitor (unregistered)
Description	Any user, including non-registered visitors, can browse active relief campaigns, filtered by disaster type or cause, organization name, or geographical region.
Pre-condition	At least one active campaign exists in the system.
Post-condition	The user is presented with a filtered, accurate list of active campaigns matching the selected criteria.

FR3 — User Login and Authentication

Actor(s)	Donor, Organization

Description	Registered entities access the system via their designated credentials through a secure login gateway. If an incorrect password is provided consecutively, the account is temporarily locked to prevent brute-force attacks, requiring identity verification to restore access.
Pre-condition	The user holds valid, previously registered credentials.
Post-condition	On success, the user is granted an authenticated session; on repeated failure, the account is temporarily locked pending identity verification.

FR4 — Execute Donation (Smart Contract Integration)

Actor(s)	Donor
Description	A registered donor can contribute funds to a verified campaign. The transaction is governed by a decentralized smart contract, ensuring funds are routed directly to the organization's wallet without intermediary friction. The system enforces basic validation (e.g., sufficient balance) before execution.
Pre-condition	The donor is authenticated, the target campaign is verified and active, and the donor's wallet holds a sufficient balance.
Post-condition	Funds are transferred to the organization's wallet via smart contract execution, and the transaction is permanently recorded on the ledger.

FR5 — Transparent Ledger Tracking

Actor(s)	Donor, Organization, Visitor
Description	Every user has access to a real-time, read-only ledger interface. Donors can track the exact flow of their funds, verifying that their contribution has successfully reached the intended organization's wallet.
Pre-condition	At least one transaction has been recorded on the blockchain.
Post-condition	The requesting user views an accurate, real-time, read-only record of fund movement, with no ability to alter it.

FR6 — Fund Disbursement and Withdrawal

Actor(s)	Organization
Description	Verified organizations can withdraw accumulated funds to their operational accounts. This withdrawal event is also written to the blockchain, ensuring end-to-end financial accountability.
Pre-condition	The organization is verified and holds a positive accumulated balance from completed donations.
Post-condition	Funds are transferred to the organization's operational account, and the withdrawal event is permanently logged on the blockchain.

4.7 Requirements Priority Ledger

Following the identification of major functional requirements (FR), identifiers and priorities are assigned below for development phasing.

ID	Requirement	Priority
R1	User and Organization Registration	**High**
R2	Secure User Login and Authentication	**High**
R3	Search and Browse Relief Campaigns	**Medium**
R4	Execute Blockchain-backed Donation	**High**
R5	Transparent Ledger / Transaction Tracking	**High**
R6	Fund Disbursement Logging	**Medium**

5. Non-Functional Requirements

To ensure the system is resilient, scalable, and secure, the following non-functional requirements (NFR) must be strictly adhered to.

5.1 Performance Requirements

- **High Availability:** As disaster relief requires immediate action, the system must maintain a 99.9% uptime, remaining accessible 24x7. Rationale: relief operations are time-critical, and any downtime directly impedes fund flow during emergencies.
- **Transaction Throughput:** The blockchain backend must support concurrent transaction processing without network bottlenecking during high-traffic emergency events. Rationale: donation volume typically spikes sharply in the first days after a disaster is publicized.

5.2 Security and Transparency Requirements

- **Cryptographic Integrity:** No user passwords or private wallet keys shall be stored in plain text. All sensitive authentication data must be cryptographically hashed (e.g., SHA-256). Rationale: protects donors and organizations from credential theft even in the event of a data breach.
- **Immutability:** Once a donation record is written to the blockchain, it cannot be modified, deleted, or obscured by any party, including the system administrators. Rationale: immutability is the foundation of the trust the system is designed to restore.

5.3 Design Constraints

- **Platform Independence:** The application must be developed using modern web standards (HTML5, CSS3, modern JavaScript frameworks) to ensure cross-browser compatibility across Google Chrome, Mozilla Firefox, Safari, and Edge.
- **Simplicity Metric:** The user interface must hide the underlying complexity of the blockchain, allowing non-technical users to donate with the same ease as a standard e-commerce transaction.

Approval

This document represents the finalized Software Requirements Specification for the Donation and Relief Fund Management System as of the version and date recorded in the Document Control section above.

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